the satisfactory operation of the devices have been due almost entirely to the fact that the railroad companies have not employed proper men to maintain them. The word "automatic" attached to signaling apparatus seems to imply to the average railroad man that it is self-sustaining as well as self-operating. This is a most serious mistake.

For example, consider the automatic rail circuit system in connection with block signaling, operating highway crossing bells, switch and bridge locks, locks on interlocking levers and a variety of other devices. The track circuit, in order that it shall be effectively established, demands that every rail joint shall be bonded with wires; that the block sections shall be separated, electrically, from their neighboring block sections by insulated joints; that the batteries shall be set up and connected by substantial conductors properly protected and connected to the rails; that the relay or electrical instruments shall be protected from harm and interference, in order that they shall be operative. The rails, joints, ties and roadbed are kept in condition for traffic by a gang of men whose chief duty this is. It is customary to require these men to look after the track attachments of signals also. They are familiar with ordinary work of the track and have more or less of a standard by which to work. They know when a rail joint needs lining, tightening or renewing. They know where, when and how spikes must be driven. Their duties require that the track shall be in good condition for the passage of trains, clearing the roadbeds of weeds, dressing ballast, renewing ties and many other incidental matters are left for such times as they are not thus employed. Bond wires, track insulations, connections to batteries and similar attachments, where such exist on a section, are constantly passing under their eyes. They have an indistinct idea of their purpose and importance and have received orders to care for them. The apparent mystery of the thing stands in the way of intelligently doing this and, together with the weeds, ballast and other things they, like the telegraph men, put off the signals for a more convenient time until matters have long gone wrong. They have their regular duties and are subject to special call orders. As a rule they are the least intelligent, worst paid employees and it is unreasonable to expect them to understand the simplest details of signaling apparatus and the accessories entering into the track department.

The various parts of signal apparatus are simple in themselves, but they each form a link in a chain, and one defective factor spoils the whole. The automatic signal never fails without reason, and a broken link due to neglect is usually the reason. The remedy lies in treating the subject with the same degree of care and watchfulness as is devoted to the other important factors which go to make up a safe railroad. Let us ask how often locomotives are inspected, and how about bridges and rock cuts along the line? We are at present limiting ourselves to crossing signal apparatus, which is but a small part of the whole department of automatic signaling, but very serious accidents may occur at crossings, and a special reason for the most careful maintenance of such devices lies in the fact that it is very difficult to prove that the apparatus worked properly at the time of an accident. Frequent inspection and constant care are the price to be paid for the best evidence that is to be had that the instruments were doing their duty, and this will also tend to render occasions for such evidence less frequent.

Staybolts Broken and Partially Broken.

Among the noon-hour topical discussions at the recent Master Mechanics' convention was one upon Broken Staybolts, which was introduced by Mr. T. A. Lawes, Superintendent of Motive Power of the C. & E. I. Railroad with the following remarks, which form the basis of another article upon the same subject presented elsewhere in this issue:

Under the presumption that you will readily grant me permission to qualify the phrase, I shall be pleased to present to you a few thoughts on the subject of "Partially Broken Staybolts," which, in my opinion, is a topic of much greater importance, since partially broken staybolts are the more difficult to detect. In fact, so far as my investigation goes, they are never detected under the old methods, and must be regarded with suspicion.

For some years hollow staybolts and drilled staybolts have been used to a limited extent, but for some reason—or no reason—neither one has been put into general use, although, as I believe, the protection afforded is invaluable.

To satisfy myself as to an inspector's ability to detect broken and partially broken staybolts, I have had the staybolts in 13 engines drilled during the past year. The plan adopted was to have the inspector locate all the broken staybolts he could find, after which the staybolts in the firebox were drilled, including those marked by the inspector as broken. In the first firebox tested in this manner the inspector found 39 broken staybolts. After drilling and testing under water pressure these were all found broken—and in addition to these we found 59 others broken which the inspector was unable to detect by the hammer test. This surprising result led me to examine the broken staybolts critically, and I found that those detected by the hammer test were broken entirely off, while those found by drilling holes in the ends were only partially broken off.

After testing the stay-bolts in 12 fireboxes and finding the ratio of broken staybolts and partially broken staybolts about the same as in the first firebox tested, I concluded to try the method of testing under boiler pressure, and having a helper hold on while the inspector gave the hammer test, but with no better results.

I desire now to direct your particular attention to the 13th and last firebox tested for broken and partially broken staybolts by the two methods. I consider it the most severe comparative test of all, from the fact that three inspectors in turn did their level best to locate broken and partially broken staybolts by the hammer test, they having been informed that the staybolts were to be drilled after they were through with their inspection. They were given all the time they required to make a careful and accurate inspection. The result was that the hammer test located four broken staybolts and the drilling test discovered 46 partially broken.

A careful record of the broken and partially broken bolts detected in 13 engines shows that 440 were discovered by the hammer test, and 619 by the drilling test.

To me these facts are conclusive evidence that partially broken staybolts cannot be detected by the hammer test, and I believe a great risk is run by either not drilling the ends or not using hollow staybolts, and that either of these precautions will prevent many boiler explosions with the usual verdict of "Cause of explosion unknown."

There have been more boiler explosions in recent years caused by broken staybolts than ever before. I attribute this to the fact that a much higher pressure is carried. Too great precaution cannot be taken to prevent loss of life and destruction of property, and my experience leads me to recommend that every staybolt used should be either hollow or drilled, making it self-detecting.

The expense of the precautions here recommended should not stand in the way of preventing boiler explosions due to defective staybolts, as the cost for drilling is only about $3.75 for a firebox containing 900 staybolts. I figure that the cost is only an apparent one.

For example: If the 619 partially broken staybolts had not been drilled and tested in the 13 fireboxes referred to above, these partially broken staybolts in time would have become broken staybolts, and the engines containing them would have to be laid up—possibly when needed badly—for the removal of broken staybolts, which would have necessitated the removal of jackets, air brakes, pumps, frame angle iron and other parts covering the staybolts, thus causing an additional expense over that, if properly attended to in the shop while undergoing repairs.

It is my practice to turn engines out of shop with staybolts beyond suspicion, and I find that I have no trouble with them afterward. I recommend to such members of the association as do not use hollow or drilled staybolts, that they drill the staybolts in a few fireboxes as described and keep a record of the number found by the hammer test and by drilling them, and I am sure that they will be surprised at the results.

The d'Auria Pumping Engine.

The problem of working steam pumps which are not fitted with heavy flywheels in a way permitting of using steam expansively has proved itself a troublesome one and the new pump shown in the accompanying engraving is brought to our attention with strong claims for the economical use of steam. The design is due to Mr. Luigi d'Auria and is described in a circular by the d'Auria Pumping Engine Co., from which the following information is taken:

Cut-off valves are used to secure the expansion of the steam in both simple and compound types. The expansion is made possible by an exceedingly simple and ingenious compensating device which adds nothing to the working parts of the pump that is at all likely to give trouble in operation. This device consists mainly of a liquid column contained in a closed loop of pipe extending from the ends of a chamber in which a piston or plunger is fitted and carried by the same rod upon which the steam piston and the

LUBRICATION OF LOCOMOTIVE CYLINDERS.

The lubrication of locomotive valves and cylinders is a subject which is very much alive just now and in the hope of adding something of value to the discussions which have already been held during the past few months these paragraphs are presented.

It has been demonstrated that engines and steam pumps will run indefinitely without oil. That running in this manner admits of saving the cost of oil is axiomatic, but because of the fact that, next to initial condensation of steam, friction causes the greatest loss of power that occurs in the steam engine, the practice which in some branches of steam engineering seems to be gaining popularity, deserves sharp criticism. According to some authorities water is a lubricant, but while the statement that this is not true may be disputed, the affirmation is made that water does not lubricate the frictional surfaces of steam engines. It does prevent cutting, but cutting is usually due to heating and water tends to prevent heating. For this reason water is often erroneously considered a lubricant. Motive power superintendents do not voluntarily omit the lubrication of valves and cylinders, but it is true they are at present running locomotives when under steam without sufficient lubrication. They are just beginning to discover the fact and are trying to find a remedy. The idea of employing sight feed lubricators is not faulty. These instruments are the fruit of a vast amount of study and experiment, and they are beautifully successful in performing their office of delivering regularly a small amount of lubricant which may be nicely controlled. The trouble is unquestionably beyond the lubricator and it lies in the method of sending the oil to the steam chest. It lies in the pipe system employed, through which the oil is expected to travel a downward sloping course from the lubricator to a point perhaps 6 or 7 feet below it to the steam chest. The oil feeds well enough when the throttle valve is closed and when it is aided in its journey by vacuum, but the difficulty begins when the throttle opens and steam at a pressure but little below the boiler pressure is admitted at the steam chest end of the pipe. The increase of working boiler pressures during the past few years has been blamed for the trouble, but from experiments recently made it appears that it exists with pressures as low as 140 and 155 pounds per square inch, and these pressures have been used for years. It is a question whether the difficulty has not existed from the beginning of the use of the "tallow pipe" in connection with automatic lubricators. The results are more noticeable and more troublesome with high pressures, which are used to assist in meeting the vastly increased duty expected of modern locomotives, but it is not believed to be true that the high pressures cause the lack of regular feeding of oil. Accompanying the use of increased pressures longer locomotive runs are now the rule, which makes it more important to lubricate properly.

In experiments made upon a Western road last year it was found that with a boiler pressure of 140 pounds and the locomotive running at a rate of 25 miles per hour with wide-open throttle and the automatic lubricator feeding 60 drops of oil per minute, the oil did not appear at the steam chests until 30 minutes had elapsed. The oil pipes had been blown out with steam before this test, and were clean and empty when the locomotive started. More cases of similar character may be cited, but it is not necessary to state them. Some interesting indicator cards, taken by Mr. E. M. Herr and published in the September, 1896, Proceedings of the Western Railway Club, show that while one cylinder of a locomotive may be developing 795 horse-power, its valve being well lubricated, the other cylinder when the lubrication is deficient may give only 589 horse-power, a difference of 206 horse-power between the two cylinders. Without changing anything about the locomotive except the lubrication, the cylinders were made to indicate equal power, the difference in the first case being due to the springing of the rods and connections of the valve motion due to the high resistance of the poorly lubricated valve. This springing prevented the valve from opening and admitting enough steam, yet the lubrication was not defective enough to cut the valve during the time covered by the experiments. The same authority states that he has found a difference of 50 per cent. between the two cylinders of a locomotive from the same cause, when the eccentric rods were long and crooked.

One of the best writers on locomotive subjects in explaining the operation of automatic lubricators seven years ago, after speaking favorably of the automatic apparatus, said, gingerly, that the lubricant "is said to reach and oil all parts of the valves and cylinders." At that time he was in doubt as to the oil reaching its intended destination. The reason why the oil does not go through the pipes regularly is interesting. It is reasonable to suppose that at the opening of the throttle steam under boiler pressure is forced up into the long oil pipes, where it condenses, nearly filling the pipes with water, and the lighter oil cannot get down through the water under these conditions. Some of it may get through on account of the churning which it receives, but it is fair to ask the question as to why the oil should be expected to descend several feet against the force of gravitation. It is well known that the old hand oilers when placed in the cab would not feed against steam pressure from the chests, and why should the automatic lubricator be expected to do so? In stationary and marine practice the automatic lubricator is usually placed directly upon the steam pipe, or steam chest, or with the connecting pipe, when one is used, so short that condensation will not interfere with the operation.

A number of suggestions have been made for overcoming the difficulty on locomotives. Each of the prominent lubricator manufacturers has applied devices for the purpose of admitting steam into the oil pipe at the lubricator at or near the boiler pressure. This is done apparently with the object of counteracting the effect of the pressure from the steam chest when the throttle is open, and the admission of the steam at the lubricator is made automatic, depending upon the opening of the throttle. Without doubt these devices assist in securing circulation in the oil pipes. Another plan which appears to be successful in getting the oil down is to apply a steam ejector at the steam chest end of the pipe to suck the oil down. All of these involve complication apparently without getting at the trouble in the most direct way, which would appear to be to prevent the lodgment of condensed steam in the piping. In marine practice it has been found necessary in using the rising drop sight feed to place the lubricator below the cylinder and to arrange the oil pipe so as to be nearly vertical, with no dipping bends, and even then it has been found necessary to use a pipe not smaller than one-half inch in diameter. The working of this device depends upon the fact that oil is lighter than water, volume for volume, and many sight feed lubricators using the rising drop have been condemned in this practice, when the only trouble was that the lubricator was placed above the cylinder level, The easiest way out of the difficulty would appear to be to employ the falling drop sight feed, since the lubricator must be placed above the level of the cylinder, and to use oil pipes of such large diameters as to prevent the accumulation of suspended water. If a pipe is full of steam there will be no tendency to hold the oil back, even if the pressure fluctuates on account of the intermittent admission through the main steam valves. There can be no objection to the use of pipes of one inch or more in internal diameter if this is found to be necessary. One pipe of this size is better than two pipes, several valves, differential pistons and other complications found necessary in the suggested devices. It is expected that such a suggestion as this will call forth criticism, but it is believed that the problem is being attacked at the wrong end.

METHODS OF APPLYING ELECTRIC MOTORS IN SHOPS.

In the application of electric motors to the driving of machine tools, three methods of procedure naturally come up for consideration. The first of these is the application of a motor to each tool, the second is the grouping of tools and the installation of a motor for each group, while the third is the driving of considerable lengths of main shafting by motors, the belting to countershafts and the arrangement of the tools being practically undisturbed. There is something to be said in favor of each of these methods, but where the sole object is to save a large part of the